Open Science in Qualitative Research

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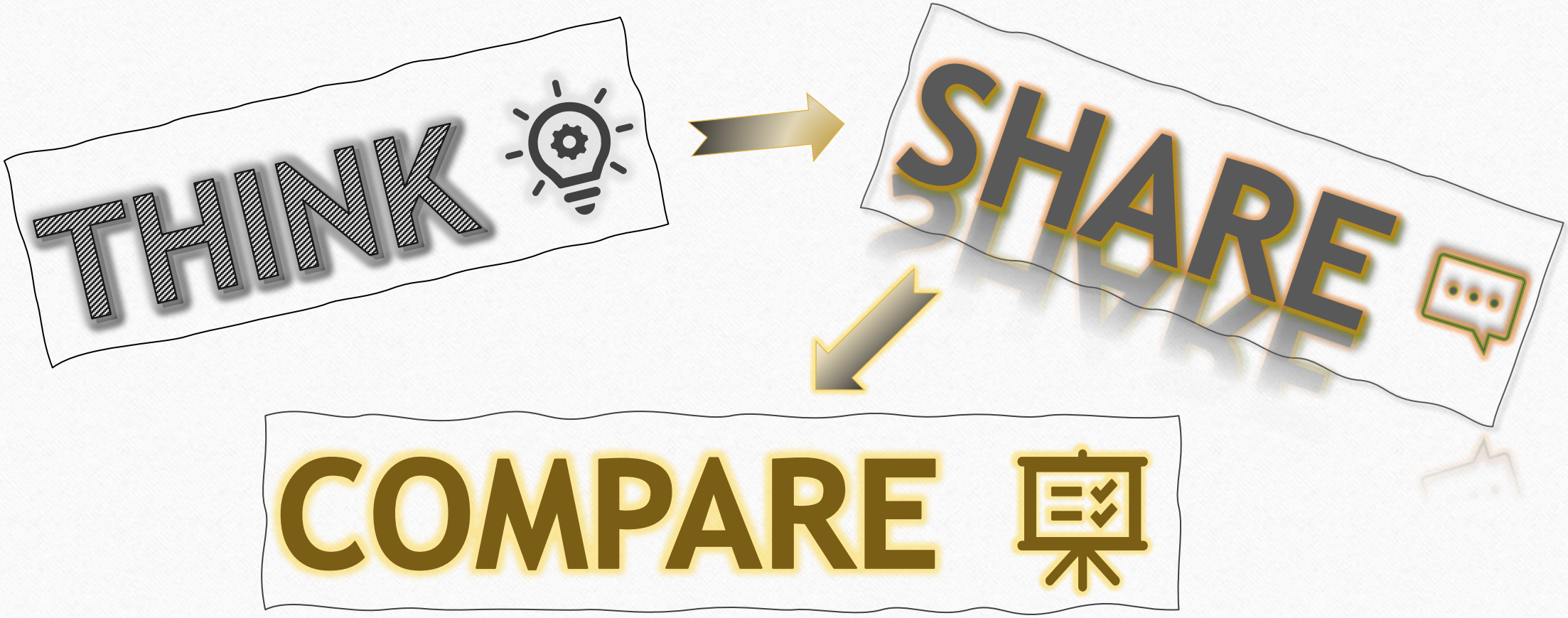
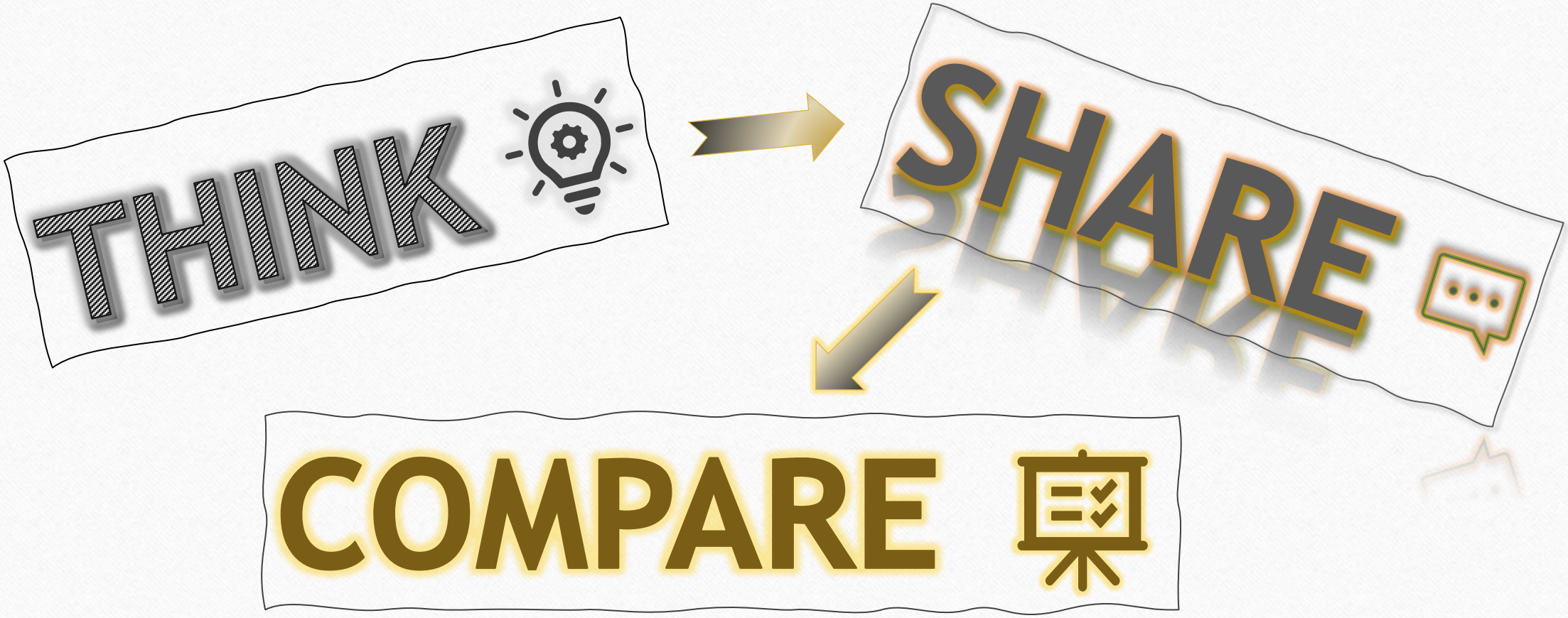
Who is here today?

To answer: Go to menti.com and use code 2323 3719



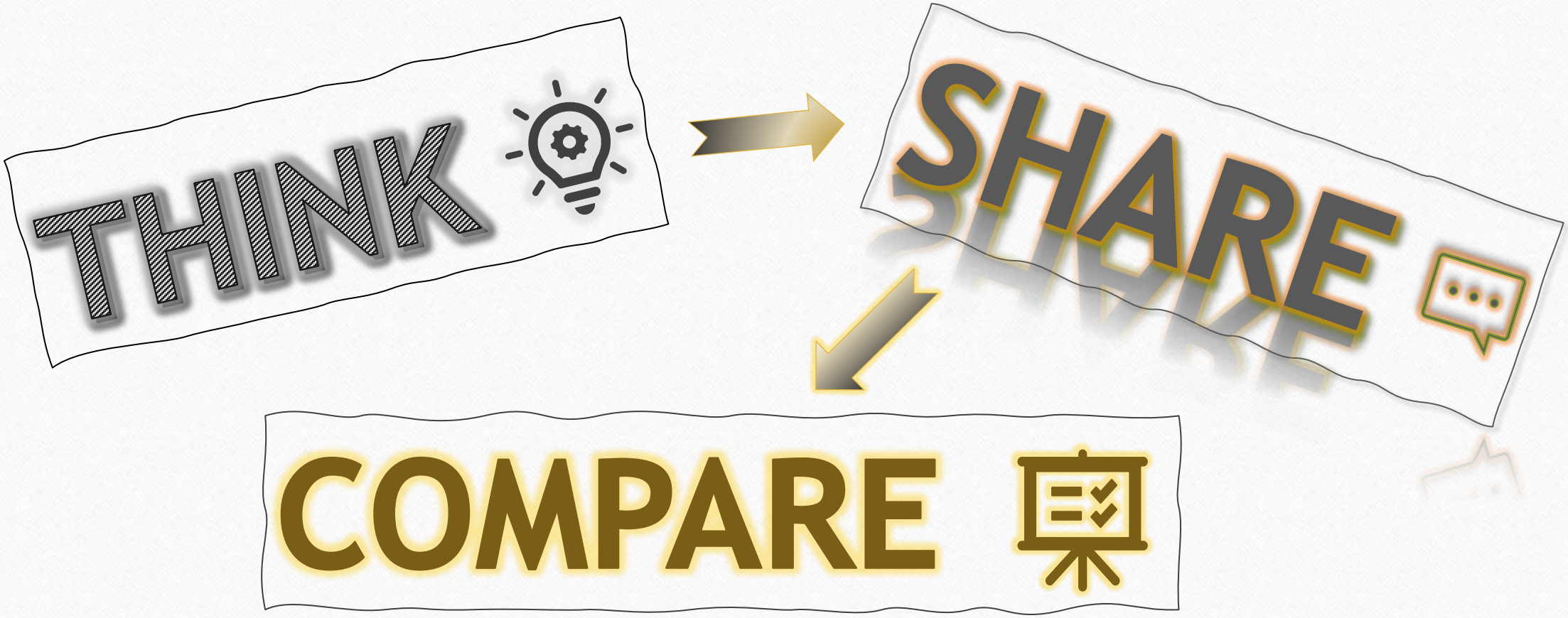
What is the first thing that comes to mind when you think of "Open Science"?

To answer: Go to menti.com and use code 2323 3719



Have you already used Open Science practices in your qualitative research?

To answer: Go to menti.com and use code 2323 3719



TODAY

1. What is Open Science?
2. Open Science practices
3. Ten steps towards Open Science
4. Making a commitment to Open Science

1. What is Open Science?

Open Science

…inclusive construct that combines various movements and practices → diversity

aiming to make multilingual scientific knowledge openly available, accessible and, reusable for everyone, → openness

to increase scientific collaborations and sharing of information for the benefits of science and society, → collaboration

and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community → transparency

Source: UNESCO (2021)

2. Open Science practices

Open Science: 13 categories of practices

1. **Pre-registering** study plans
   - Dedicated qualitative templates (see Additional Resources slide)
   - Guaranteed publication with **registered reports** (Karhulahti, 2022b; Karhulahti et al., 2023)
2. **Open Collaboration**: Publicly sharing project plans to encourage feedback and collaboration
   - On dedicated platforms such as the Open Science Framework www.OSF.io or www.Academia.edu
   - Look out for and engage with shared project plans

Adapted from: Schneider (2024)

Open Science: 13 categories of practices

3. **Open Data**: Deciding when, how, and with whom to share research data

- Journals and funders increasingly expect FAIR data sharing (Prosser et al., 2022)
  - **FAIR** data: **F**indable, **A**ccessible, **I**nteroperable, **R**e-usable (Jones et al., 2018)
- Can and should we share qualitative data (*inter alia* Branney et al., 2019; Joyce et al., 2022; Karhulahti, 2022a)?
- Planning to share your data? Include this in your ethics application and ask your participants for permission via the informed consent form
- Sharing your data? Deposit them in a data repository/archive to get a DOI

Adapted from: Schneider (2024)

Levels of access to research data

Level of Access	Definition
Open	Data are freely available for use in accordance with general use agreement of repository and standard citation practices
Restricted	Data are available for use when the user meets standard criteria set by data repository to ensure ethical use of data (could include obtaining ethical advice from an ethics committee or accessing the data through a virtual environment)
Controlled	Data are available for use when the user is approved by the original researcher (access could depend on research questions and intended analysis; access method and amount of data shared are decided by the original researcher)
Closed	Data deposit and citation exist for archival purposes, but no data are currently available (could be embargoed until publication of results, change in sensitive situation, death of a participants, or a certain duration of time from collection)

Adapted from Jones et al. (2018)

Open Science: 13 categories of practices

4. Open Materials: Publicly sharing research materials

- Research instruments such as interview and focus group schedules
- Stimuli and discussion prompts

5. Open Methodology: Publicly sharing (fragments of) the research methodology

- Annotation or category schemes
- Annotated or coded transcripts
- Codebooks
- Memos

Adapted from: Schneider (2024)

Open Science: 13 categories of practices

6. Citizen Science: Involving the non-academic public in the research process

- Citizens are not just sources of data, but can also contribute actively, creatively, and meaningfully to data generation and analysis
- Zooniverse: platform hosting projects that employ Citizen Science

7. Using **Open File Formats**

- Using data analysis software that generates accessible and re-usable files
- Tip: Preferred open file formats differ across institutions -> check the recommendations of the repository where you're planning to deposit your work

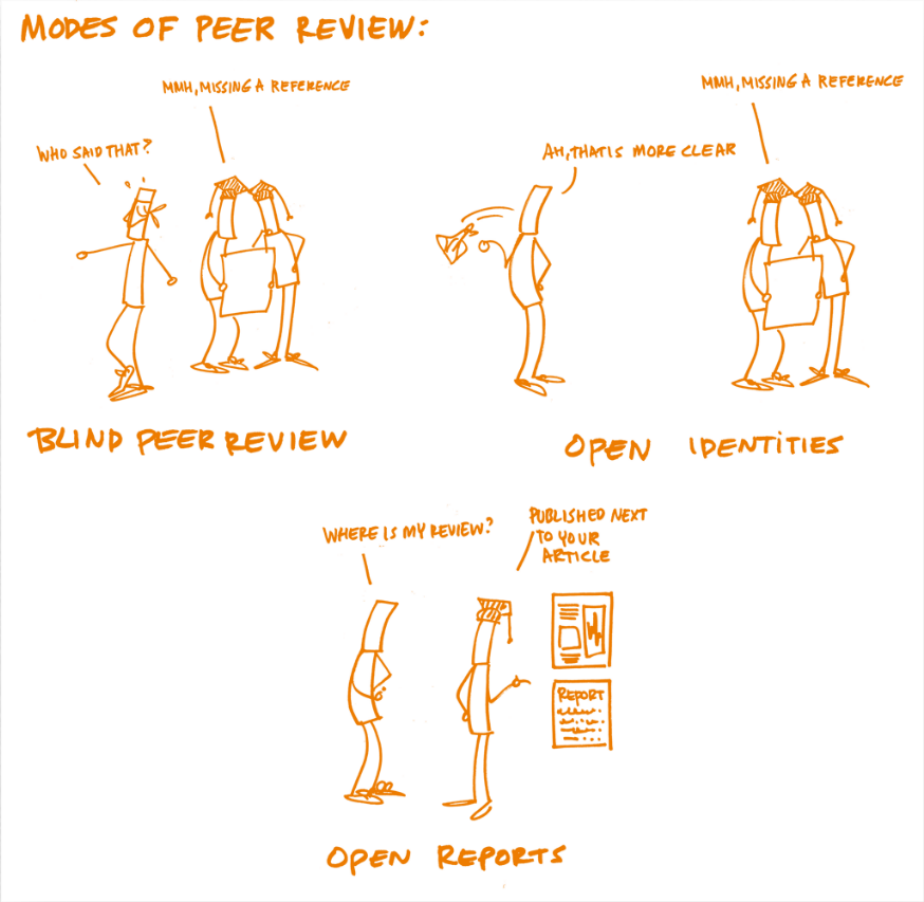
Adapted from: Schneider (2024)

Open Science: 13 categories of practices

8. **Open Code / Open Script:** Publicly sharing data analyses
   - Published scripts (code) produced during data analysis
9. Providing **open-source** code of software
   - Software produced as part of a research project is made available for reuse, adaptation, and further distribution by publishing it with an open-source license

Adapted from: Schneider (2024)

Open Science: 13 categories of practices



10. **Open Peer Review:** Deciding for openness in the peer review process

- Default: double-blinded peer review
- Revealing the identity of the authors and/or reviewers
- Publishing review reports alongside the article
- Publishing authors' responses to reviewers' comments alongside the article

Adapted from: Schneider (2024)

Open Science: 13 categories of practices

11. Publishing **Open Access**
    - Publishing research articles online with no access or re-use fee and restrictions
    - Green open access: authors upload the **author accepted manuscript** to their university's research repository as soon as it has been accepted
    - Gold open access: journals make the **final (typeset) version** of the article available free of charge for readers, after the authors have paid the article processing charges (APCs). APCs can be covered by universities or research grants
    - Platinum/diamond open access: journals make the **final (typeset) version** of the article available free of charge for readers without any APCs
    - When and which version of your manuscript can you upload to ResearchGate, Academia.edu etc? Check on SherpaRomeo

Adapted from: Schneider (2024)

Open Science: 13 categories of practices

12. **Science Communication**: Communicating research results to nonacademics
    - In person (e.g., science festivals) or online (e.g., blogs)
    - In writing (e.g., De Focus) or in video/audio form (e.g., podcasts)
    - To lay audiences or to interested groups (e.g., professionals)

13. Generating **Open Educational Resources**
    - Massive Open Online Courses (MOOCs)
    - YouTube /TikTok tutorials
    - Handbooks
    - Guidelines

Adapted from: Schneider (2024)

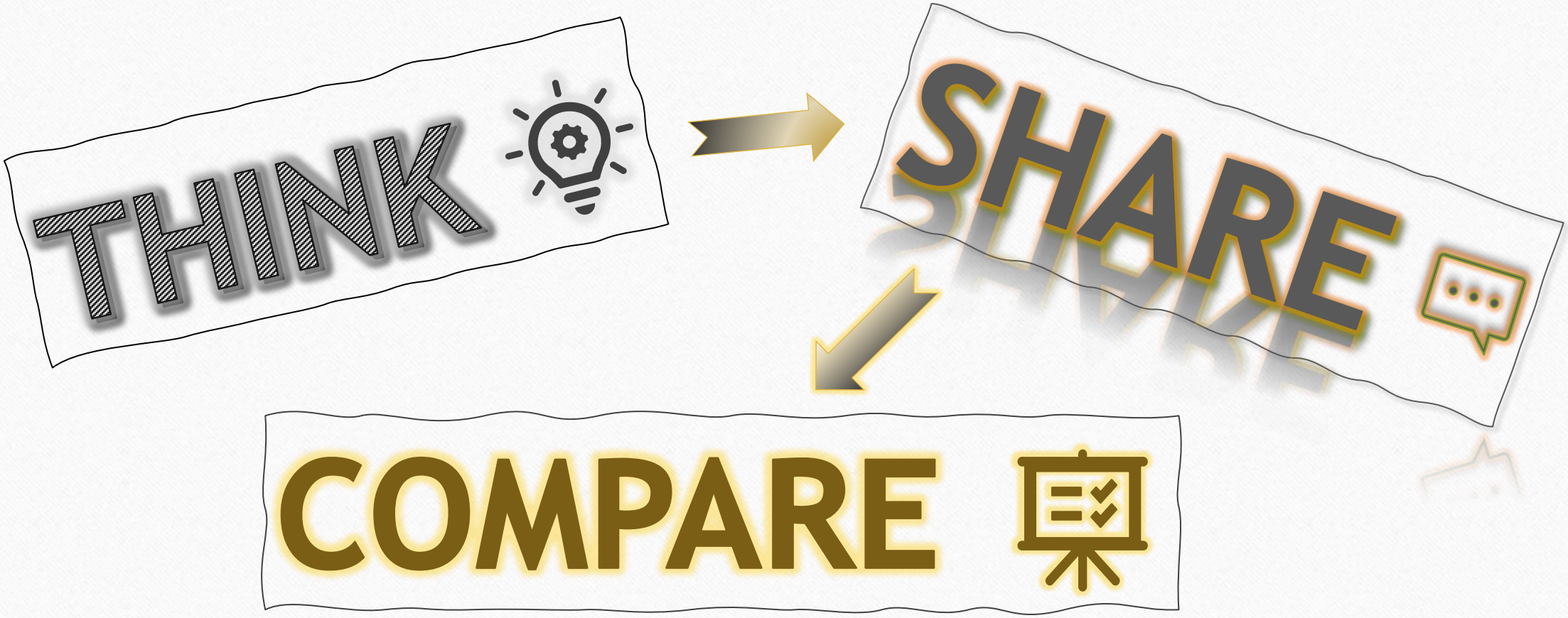
Open Science-flavoured Lunch Break

- You received a €1000 award to implement Open Science practices in your current research project. What would you spend the money on?
- Use Padlet (accessible via the QR code) to share your answers

Take
a
Break

If you had to implement Open Science practices in your research, what would be some of the barriers and challenges you would face?

Discuss in pairs and then share your conclusions with the whole group



3. Ten steps towards Open Science

Adopting Open Science practices: Where to start?

1. Join the ReproducibiliTEA Amsterdam journal club and the Open Science Community Amsterdam (OSCA)
   - Tip: Sign up to the OSCA newsletter for OS updates
2. Create an OSF (Open Science Framework) account
   - Upload and organise research materials (e.g., consent forms, data, analysis plans, etc.)
   - Get timestamped versions of materials and data
   - Selective access options (i.e., you decide who has access to what)

Adapted from: Branney et al. (2023) & Kathawalla et al. (2021)

Adopting Open Science practices: Where to start?

3. Pre-register your studies

- Use a pre-registration template for qualitative research (see Additional Resources slide)
- Publish your pre-registration via, for example, the AsPredicted or Research Registry portals

4. Adopt a contributorship model when writing papers

- Discuss roles and responsibilities openly with everyone in your team at several stages of the research process
- Use the Contributor Role Taxonomy (CRediT) to describe contributors' roles

Adapted from: Branney et al. (2023) & Kathawalla et al. (2021)

Adopting Open Science practices: Where to start?

5. Share your data

- Consult the excellent DANS guide on "Making Qualitative Data Reusable"
- Design ethical consent forms that allow data sharing
- Upload your data to a data repository (see Additional resources slide)
- Decide on the level of access (see Levels of access to research data slide) and what you will share (e.g., "raw" data, pseudonymised data, meta-data, etc.)

6. If applicable to your methodological approach, include a positionality statement in your article

- Deliberately reflect on how your identities, experiences, and standpoint influence your methodological decisions: topic selection, research questions, participant (data sources) selection, data analysis and interpretation

Adapted from: Branney et al. (2023) & Kathawalla et al. (2021)

Adopting Open Science practices: Where to start?

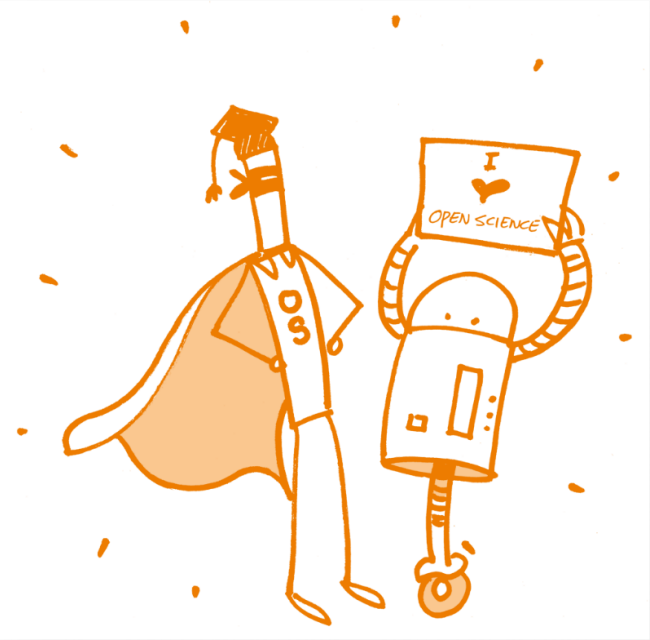
7. Upload your submitted papers to a pre-print repository
    - PsyArXiv: psychology repository
    - SocArXiv: social sciences repository
    - MedRXiv: health sciences repository
    - Zenodo: multidisciplinary repository
8. Publish your papers open access
    - Green OA: Upload your papers to the Pure research portal (details here)
9. Communicate about your research to non-academic audiences
    - Write a popular science piece
    - Record a video and post it on social media
    - Sign up to take part in a science festival / event
10. Find out more about Open Science
    - Check out the Additional Resources slide at the end of this presentation

Adapted from: Branney et al. (2023) & Kathawalla et al. (2021)

4. Making a commitment to Open Science

Beyond this workshop: Open Science Action Plan

- Pick a person in this room as your OS ‘accountability buddy’ and exchange contact details.
- Individually, make an action plan for engaging with OS in your current research project.
  - Select at least one and at most three OS steps that you would like to take in the next six months
  - Describe in as much detail as you can how, when, and why you will adopt these practices
  - Think about possible hurdles in adopting these practices. How will you pre-empt and overcome them?
- Write down your answers.



- Make an appointment to share and discuss your answers with your OS ‘accountability buddy’.
- Optionally: share and discuss your answers with other in your research group.

Beyond this workshop: Qualitative Open Science Symposium

Qualitative Open Science: Challenges, Opportunities, Tensions, and Synergies

- When: 28th of March 2025
- Where: Vrije Universiteit Amsterdam
- Why?
  - 12 guest speakers from the UK, Norway, Germany, Austria and the Netherlands
  - 7 interactive workshops to choose from
  - Networking, community building, and collaboration opportunities
  - Free to attend and no preparations required in advance
- Places are limited!

Check out the full programme & sign up to attend the event

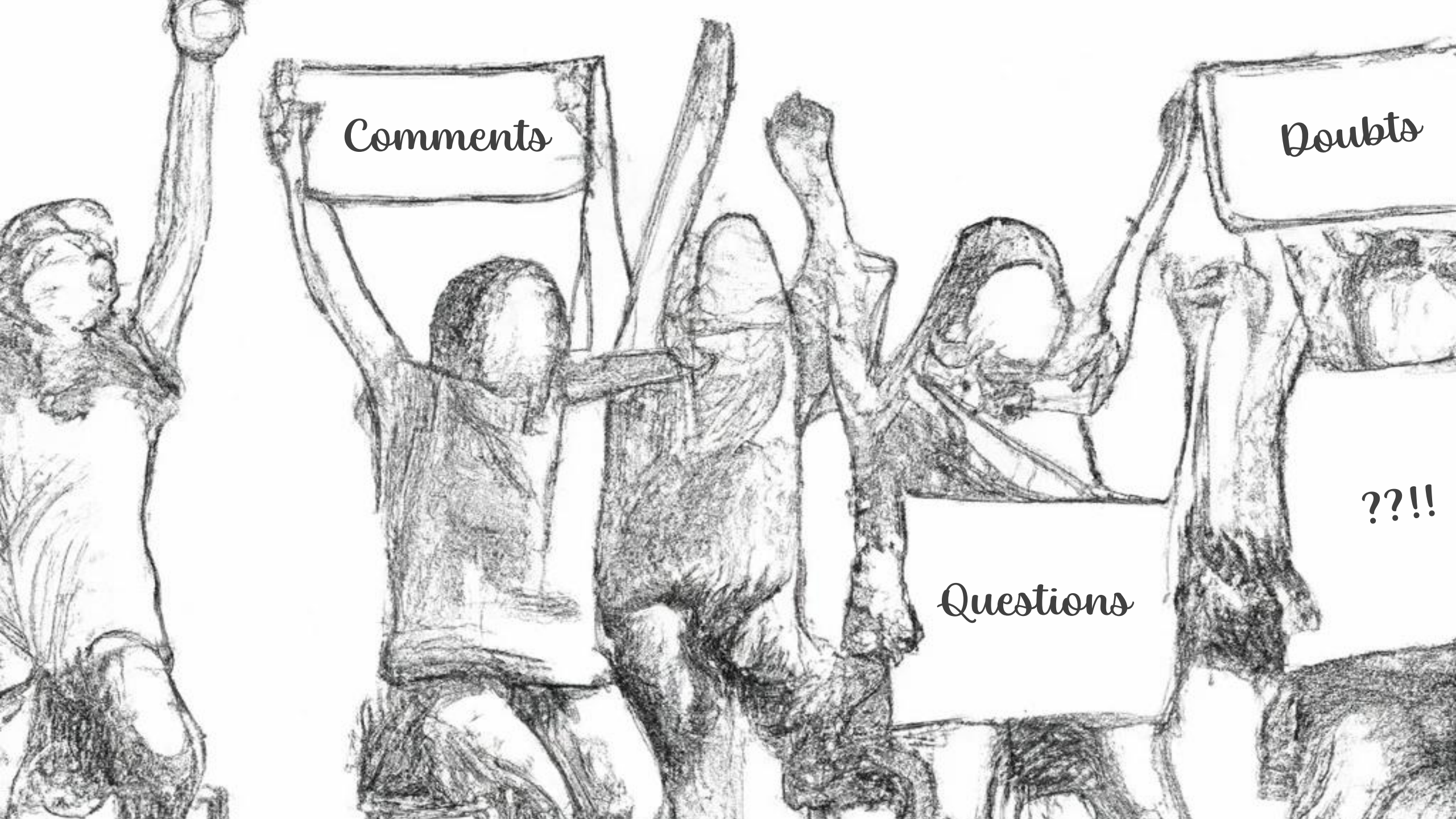

Comments
Doubts
??!!
Questions

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Image sources

- Cover slide image was created with the assistance of DALL·E 2
- “Take a break” slide image was adapted from the photo by Alessio Zaccaria on Unsplash
- Open Peer Review image was sourced from https://www.fosteropenscience.eu/sites/default/files/Cartoons.zip
- Open Science Action Plan image was sourced from https://www.fosteropenscience.eu/sites/default/files/Cartoons.zip
- Q&A slide image was created with the assistance of DALL·E 2

Additional resources

Readings

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Additional resources

Open Science information and training resources

The FOSTER portal https://www.fosteropenscience.eu/

The Framework for Open and Reproducible Research Training https://forrt.org/

Repositories for sharing qualitative data

DANS Data Station Social Sciences and Humanities https://ssh.datastations.nl/

UK Data Service https://ukdataservice.ac.uk/

The Qualitative Data repository https://qdr.syr.edu/

Open Access resources

List of pre-print repositories: https://en.wikipedia.org/wiki/List_of_preprint_repositories

Open Science portals

The Open Science Framework (OSF) https://osf.io/

Pre-registration resources

As Predicted https://aspredicted.org/

Templates for pre-registering qualitative research: https://osf.io/dv8km/ and https://osf.io/j7ghv/